

Detecting changes in mixed-sampling rate data sequences

Joint work with Aaron Lowther and Idris Eckley

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Cardiff Nov 2022

Two indefinite lectureships (possibly more) are available at Lancaster

We are looking for people who enjoy doing methodological research in any area of statistics (broadly interpreted) and are interested in application motivated research. Happy to chat to anyone who is interested.

Deadline: 4th December

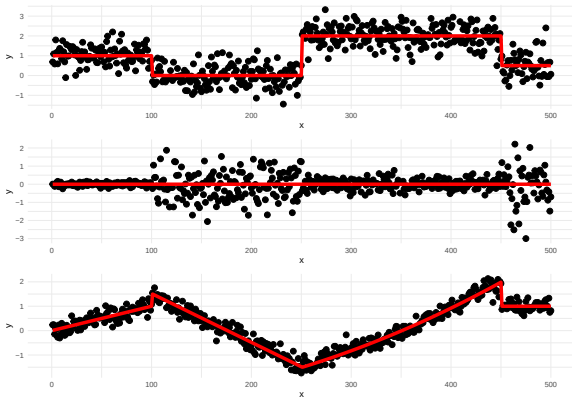
<https://hr-jobs.lancs.ac.uk/Vacancy.aspx?ref=1089-22>

- Motivation
- Method
- Simulations
- Application

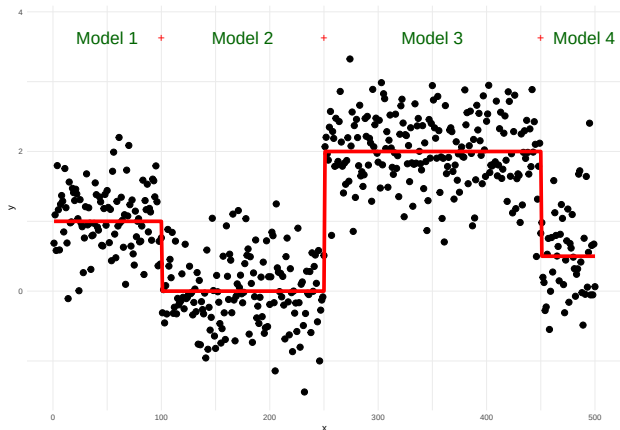
What are changepoints?

For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

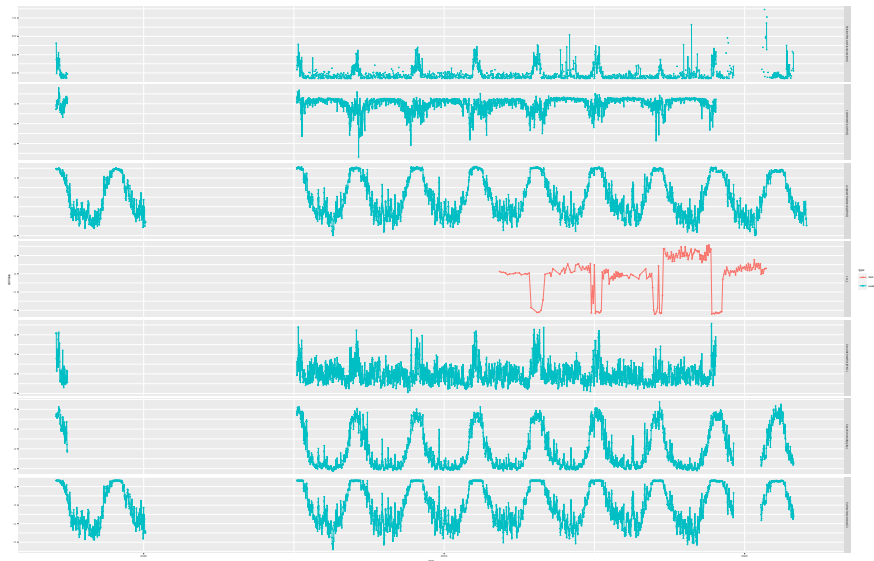
There are many different types of change.



Problem



Motivation Ice



Combine data collected using different sampling rates to detect changes in all series

- No subsampling / aggregation
- No interpolation

Problem: Existing techniques assume

- regular sampling rates
- observations across variables are observed at the same time

Key points:

- Likelihood doesn't assume regular sampling rate
- Create an observed time index across variables
- Use all data available up to a point to detect change
- Respect minimum segment lengths per series
- Respect penalties grow with sample size

Usually see:

$$\mathcal{C}(\mathbf{y}_{u:v}) = \sum_{p=1}^P \mathcal{C}_p(y_{p,u:v}),$$

where $u : v = \{u + 1, u + 2, \dots, v\}$.

Now we have:

$$\mathcal{C}(\mathbf{y}_{(u,v]}) = \sum_{p=1}^P \mathcal{C}_p(y_{p,(u,v]}),$$

where $(u, v] = \{t \in (u, v]\}$.

$\mathcal{C}(y_{p,(u,v]})$ may not be defined if $|y_{p,(u,v]}| < L_p$.

Naive solution: restrict to feasible locations.

Instead:

$$\tilde{\mathcal{C}}_p(y_{p,(u,v]}) = \begin{cases} \mathcal{C}_p(y_{p,(u,v]}) & \text{if } |y_{p,(u,v]}| \geq L_p \\ 0 & \text{otherwise} \end{cases} \quad \text{for } p = 1, \dots, P.$$

$$\min_{\tilde{\tau}, M} \left\{ \sum_{m=1}^{M+1} \left[\tilde{\mathcal{C}} \left(\mathbf{y}_{(\tau_{m-1}, \tau_m)} \right) + \beta \right] \right\},$$

-

$\tilde{\tau} = \{[\tau_0, \tau_1, \dots, \tau_{M+1}] : \exists p \in \{1, \dots, P\} \text{ where } |y_{p,(\tau_m, \tau_{m+1})}| \geq L_p$
for $p = 1, \dots, P$ and $m = 0, \dots, M\}.$

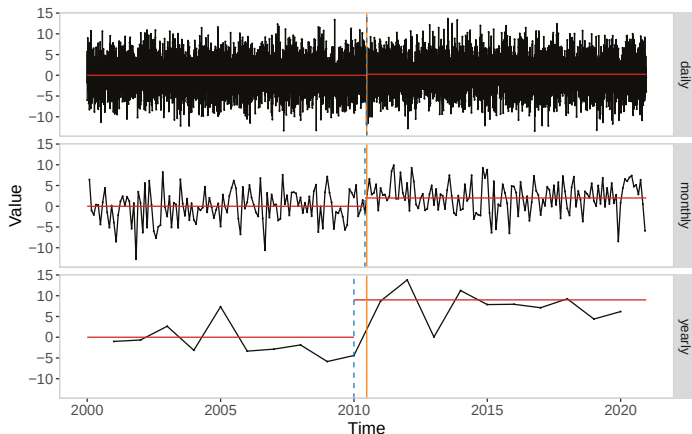
- $\tilde{\mathcal{C}}(\cdot) = \sum_{p=1}^P \tilde{\mathcal{C}}_p(\cdot)$
- $\beta = \sum_{p=1}^P \beta_p$

Solve using PELT. For $t > s$,

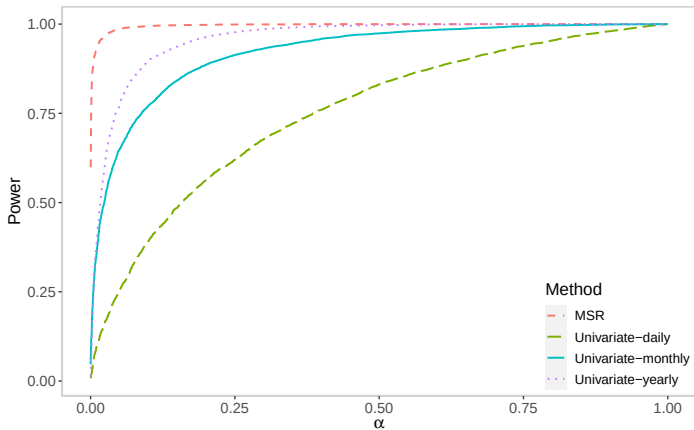
$$F(t) = \min_s \left\{ F(s) + \tilde{C}(\mathbf{y}_{(s,t]}) + \beta \right\}.$$

Pruning the set s akin to Killick et al. (2012) remains valid for the new sets.

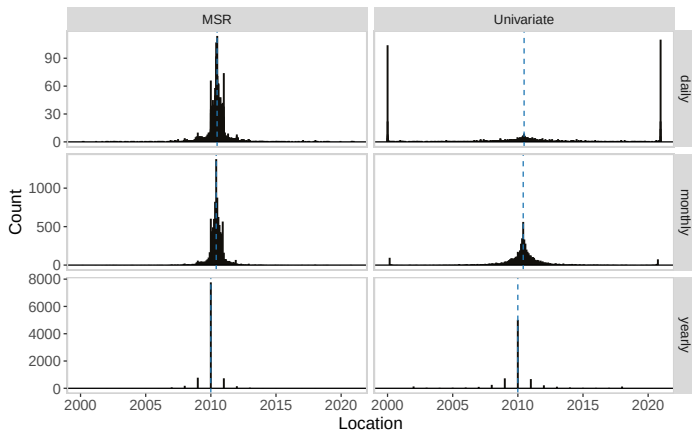
Daily-Monthly-Yearly observations. $\sigma^2 = 4$, $\mu_1 = (0.25, 2, 9)$.



Daily-Monthly-Yearly observations. $\sigma^2 = 4$, $\mu_1 = (0.25, 2, 9)$.

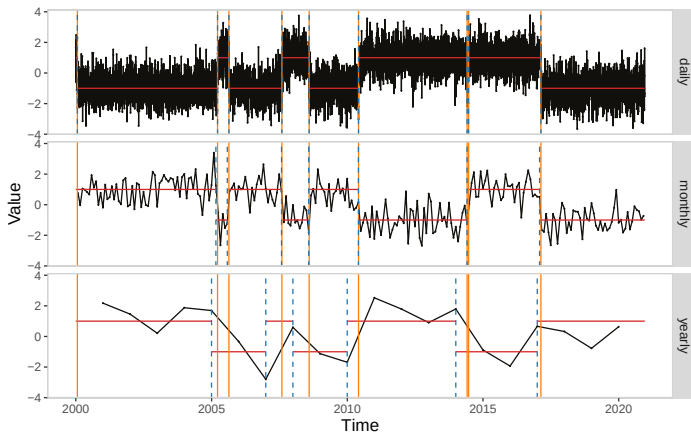


Daily-Monthly-Yearly observations. $\sigma^2 = 4$, $\mu_1 = (0.25, 2, 9)$.



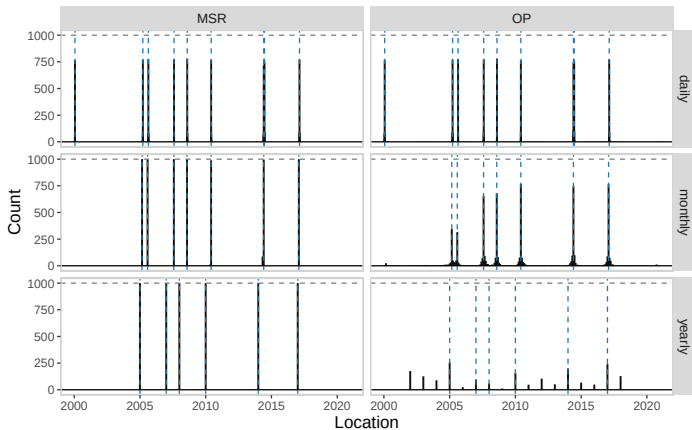
Multiple Changes

Daily-Monthly-Yearly observations. $\sigma^2 = 4$, Alternating mean (1, -1).

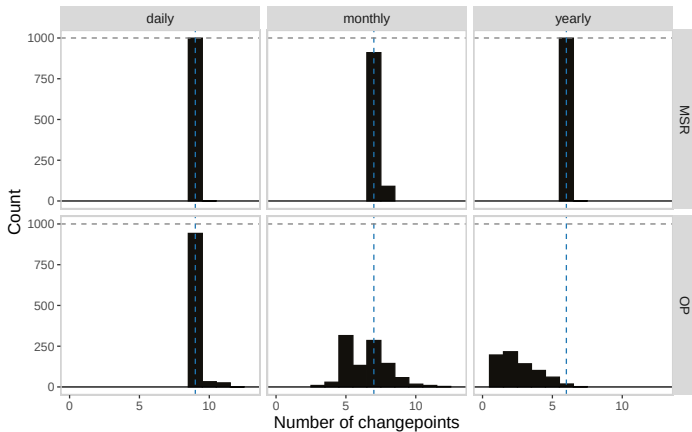


Multiple Changes

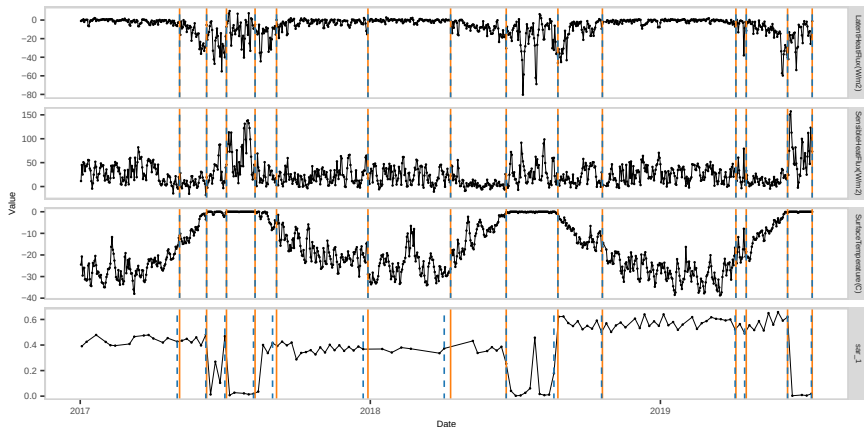
Daily-Monthly-Yearly observations. $\sigma^2 = 4$, Alternating mean $(1, -1)$.



Daily-Monthly-Yearly observations. $\sigma^2 = 4$, Alternating mean $(1, -1)$.



Application



- Able to detect changes in multivariate sequences with mixed sampling rates
- Also handles missing data within sequences
- Generic method that could be used with other test statistics

Next steps:

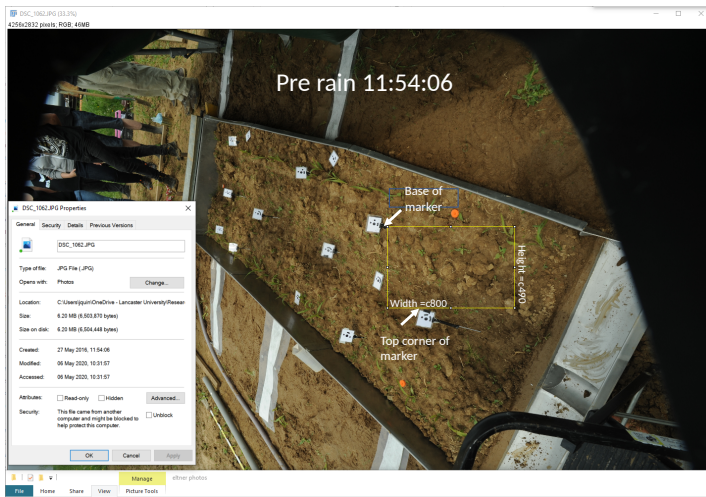
- Incorporate temporal dependence
- Incorporate cross-correlation

Detecting changes in covariance via random matrix theory

Joint work with Sean Ryan

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Cardiff Nov 2022

- Motivation
- Inspiration
- F-matrix solution
- Simulations
- Application

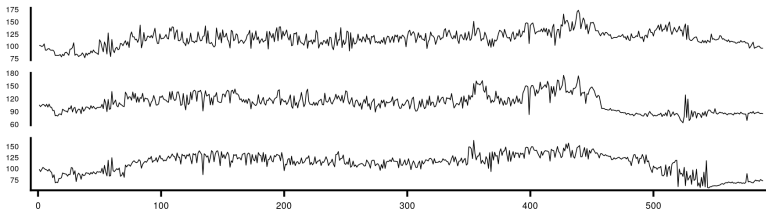


Motivation

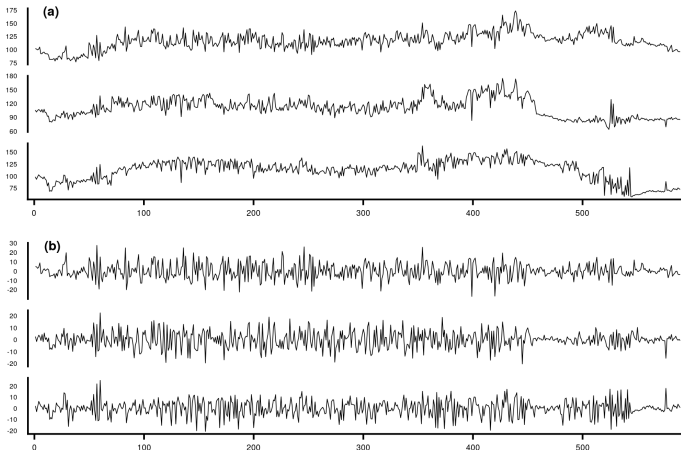


Motivation





Bayesian Trend Filter with Dynamic Shrinkage (Kowal et al. 2019), DSP R package



Detect multiple changes in covariance structure.

$$H_0 : \Sigma_0^* = \Sigma_{1,p} = \cdots = \Sigma_{n,p}$$

$$H_1 : \Sigma_1^* = \Sigma_{1,p} = \cdots = \Sigma_{\tau,p} \neq \Sigma_{\tau+1,p} = \cdots = \Sigma_{n,p} = \Sigma_2^*,$$

where τ is unknown.

Typically test statistics,

$$\psi_{\tau,1} \bar{\Sigma}(0, \tau) - \psi_{\tau,2} \bar{\Sigma}(\tau, n),$$

where $\{\psi_{\tau,1}\}_{\tau=\ell+1}^{n-\ell}$ and

$\{\psi_{\tau,2}\}_{\tau=\ell+1}^{n-\ell}$ are sequences of normalizing constants.

Large spectral norm $\implies$ changepoint.

Aue et al (2009), Galeano and Pena (2007), Wang et al (2017) . . .

All need estimate of Σ_0 for thresholds!

Univariate: use functions of $\frac{\sigma_1}{\sigma_2}$.

Advantages:

- More natural view, 1 under the null, deviations suggest changes
- No need to estimate the variance under the null
- Scale invariant

$$R(A, B) := B^{-1}A,$$

where $A, B \in \mathbb{R}^{p \times p}$.

$$T(A, B) = \sum_{j=1}^p (1 - \lambda_j(R(A, B)))^2 + \left(1 - \lambda_j^{-1}(R(A, B))\right)^2,$$

where $\lambda_j(R(A, B))$ is the j th largest eigenvalue of the matrix $R(A, B)$.

$$T(\Sigma_1, \Sigma_2) = \sum_{j=1}^p \left(1 - \lambda_j(R(\Sigma_1, \Sigma_2))\right)^2 + \left(1 - \lambda_j^{-1}(R(\Sigma_1, \Sigma_2))\right)^2,$$

- T is symmetric i.e. $T(\Sigma_1, \Sigma_2) = T(\Sigma_2, \Sigma_1)$;
- T is symmetric with respect to inversion of matrices i.e. $T(\Sigma_1, \Sigma_2) = T(\Sigma_1^{-1}, \Sigma_2^{-1})$;
- T does not depend on the underlying covariance.

If $\Sigma_1 = \Sigma_0 Z_1^T Z_1 \Sigma_0$ and $\Sigma_2 = \Sigma_0 Z_2^T Z_2 \Sigma_0$, then
 $T(\Sigma_1, \Sigma_2) = T(Z_1^T Z_1, Z_2^T Z_2)$.

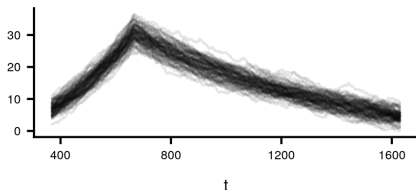
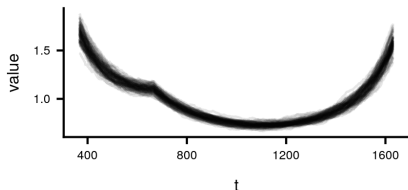
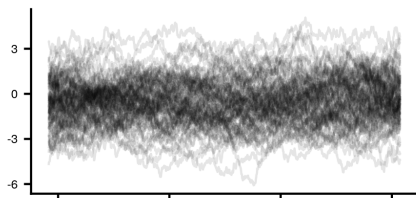
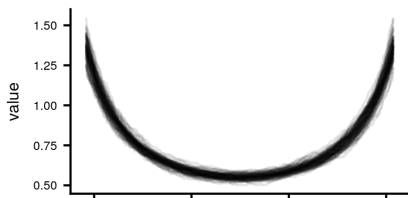
The sample sizes n_1, n_2 , and the dimension p grow to infinity such that

$$\gamma_{n_1} := \frac{p}{n_1} \rightarrow \gamma_1 \in (0, 1), \gamma_{n_2} := \frac{p}{n_2} \rightarrow \gamma_2 \in (0, 1) \text{ and } \gamma := (\gamma_1, \gamma_2).$$

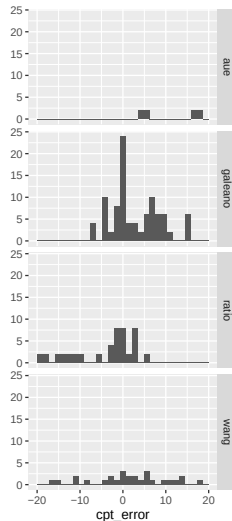
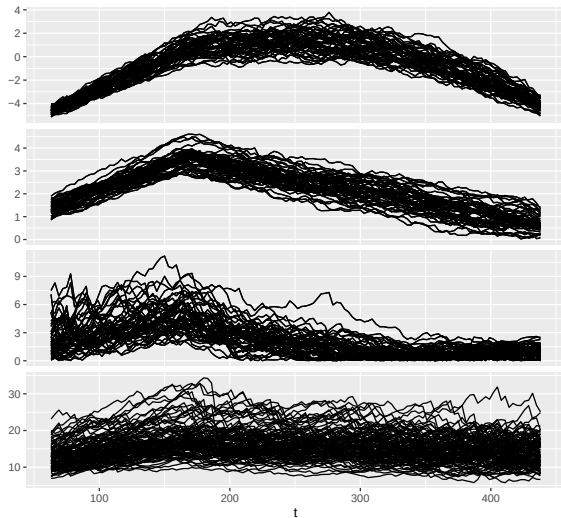
Then under some mild assumptions:

$$T \left(\frac{1}{n_1} X_{n_1,p}^T X_{n_1,p}, \frac{1}{n_2} X_{n_2,p}^T X_{n_2,p} \right) - p \int f^*(x) dF_\gamma(x) \rightarrow N(\mu(\gamma), \sigma^2(\gamma))$$

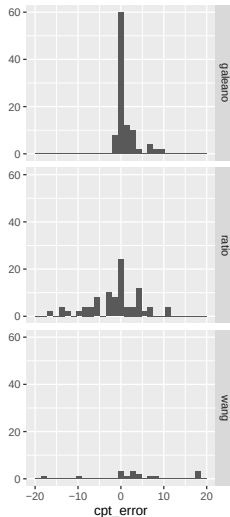
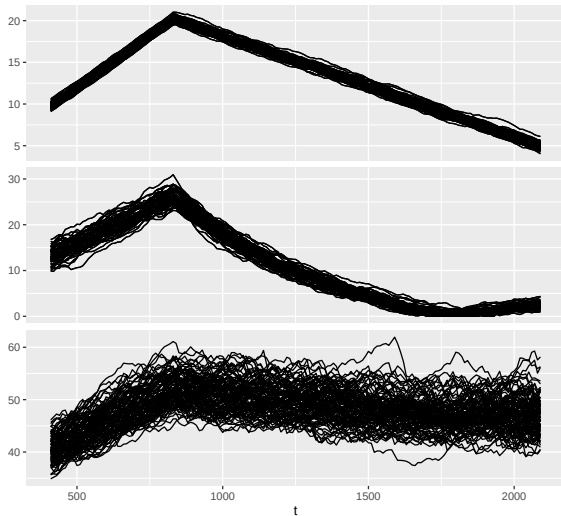
$$\tilde{T}(t) := \sigma^{-1/2}(\gamma_{t/n}) \left(T(\bar{\Sigma}(0, t), \bar{\Sigma}(t, n)) - p \int f^*(x) dF_{\gamma_{t/n}} - \mu(\gamma_{\tau/n}) \right),$$



Single changepoint small



Single changepoint large



Multiple changepoints

p	n	FPR			
		Aue	Ratio	Wang	Galeano
3	500	0.36	0.24	0.63	0.00
	1000	0.48	0.28	0.77	0.02
	2000	0.54	0.31	0.85	0.09
	5000	0.59	0.31	0.90	0.23
10	500	0.25	0.16	0.27	0.00
	1000	0.32	0.13	0.43	0.04
	2000	0.34	0.10	0.53	0.09
	5000	0.37	0.09	0.62	0.21
30	2000		0.02	0.32	0.09
	5000		0.02	0.31	0.22
100	5000		0.00	0.45	0.26

Multiple changepoints

p	n	TPR			
		Aue	Ratio	Wang	Galeano
3	500	0.55	0.38	0.51	0.00
	1000	0.59	0.48	0.48	0.01
	2000	0.63	0.60	0.44	0.03
	5000	0.65	0.68	0.45	0.06
10	500	0.56	0.55	0.51	0.01
	1000	0.69	0.75	0.54	0.01
	2000	0.76	0.87	0.58	0.04
	5000	0.80	0.91	0.61	0.07
30	2000		0.98	0.45	0.05
	5000		0.99	0.56	0.10
100	5000		1.00	0.31	0.13

Multiple changepoints

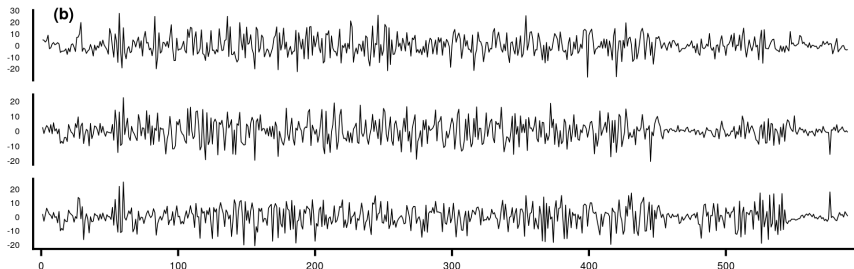
p	n	MAE			
		Aue	Ratio	Wang	Galeano
3	500	26.05	26.52	42.82	55.67
	1000	18.44	15.36	36.81	48.97
	2000	12.78	7.65	32.14	43.95
	5000	7.95	2.93	23.85	36.66
10	500	304.00	303.63	316.80	780.50
	1000	167.59	127.71	243.70	678.34
	2000	99.52	46.44	186.26	594.56
	5000	58.09	16.62	123.34	492.97
30	2000		143.43	1078.25	3079.91
	5000		51.37	525.14	2357.40
100	5000		199.51	7769.79	16281.43



Ratio: 64, 125, 184, 255, 340, 450, 527

Wang: 445 (huge eigenvalue affecting threshold)

Scientists: 64 dry - 76 very wet, increase 350, decrease 450



- New method based on ratio of covariance matrices
- Doesn't require estimation of Σ_0
- Good false positive control away from the boundaries
- Theoretical threshold works well for multiple changepoints

Future work

- Extend to mixed sampling rates and/or missingness